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# Random bipartite social networks with group effect

JunYi Wu and BangYe Wu
CSIE, National Chung Cheng University, Taiwan, R.O.C.
wcy98m@cs.ccu.edu.tw and bangye@cs.ccu.edu.tw

**Abstract**

*Generating random graphs is an important issue in social network analysis. In this paper, we propose a new generative model for random bipartite social networks in which group effect is considered. Several scenarios of network growth are considered. Different from previous models in the literatures, we focus on group effect. The simulation results show that there are significant differences between grouping and ungrouping cases. The new model also explains an interesting phenomenon in real social networks that a famous member can pull up the popularities of the ones with less popularity in his/her group.*

**Keyword:** social network, bipartite graph, power law, random graph

## 1 Introduction

A social network is a social structure made up of individuals (or organizations) called nodes, which are connected by one or more specific types of relationships. It is applied in a variety of fields such as economics, geography, information science, biology, or communication studies.

There are two types of social networks, one-mode and two-mode. For a one-mode network, there are one set of actors and kinds of relations between actors. For a two-mode network, there are two sets of actors and the relations only exist between of nodes of different sets [8]. A two-mode network can be viewed as a bipartite graph and the nodes in the two sets are called as *top* and *bottom* nodes, respectively.

There are mainly three kinds of study issues in social network analysis [4].

- Finding the social network: Empirical studies of networks probe network structure using a variety of techniques such as interviews, questionaires, direct observation of individuals, use of archival records, and "ego-centred" studies. The goal of such studies is to create a picture of the connections between individuals.
- Analyzing the network: Who are the most central members of a network and who are the most peripheral? Which people have the most influence over others? Does the community break down into smaller groups and if so what are they? Which connections are most crucial to the functioning of a group?
- Creating models: Modeling work of this type allows us to make predictions about the behavior of a community as a function of the parameters affecting the system.

Therefore, it is important to develop random graph models. Random graph models are divided into two kinds [3].

- The configuration model. For a specified features, we can generate a random graphs satisfying the required features such as degree distribution.
- The generative model. It models that posit a quantitative mechanism or mechanisms by which a network forms, usually in an effort explain how the observed structure of the network arises.

In the literatures, several models have been proposed. In this paper, we focus on bipartite graphs.

In previous works for random bipartite graphs(RBG), all top nodes are identical. But in many application, there are relations between top nodes. Usually, they are grouped with others. For example, in BBS(Bulletin Board System) network, authors and boards can be viewed as bottom nodes and top nodes, respectively. Boards are classified into classification and the behavior between top and bottom can result in this kind of bipartite social network. We propose a new

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model call grouped-RBG in which the top nodes are partitioned into several groups to exhibit the difference of grouped-RBG.

This paper is organized as follows. In Section 2, we give some definitions and notations. In Section 3, we propose the new model grouped-RBG. In Section 4, we present experimental results and discussions. In Section 5, we make some conclusions.

## 2 Preliminaries

### 2.1 Previous works and scenarios

We denote a non-weighted bipartite graph $G = (T \cup B, E)$ where $T$ and $B$ are two disjoint sets of nodes, respectively representing the top and bottom nodes, and $E \subseteq T \times B$ is the set of edges of $G$. We describe some basic models of random graphs.

- Barabasi Albert(BA) model: BA model is a generative model that use a preferential attachment principle to generate a random scale-free network. Nodes are added one by one and linked to preexisting nodes with a probability depend on the degree of these nodes [5, 6, 7, 1].
- Uniform model: When a bottom node entered the network, it randomly picks a top node to link with uniform probability and each link is independent.

BBS is an example of real bipartite social network. Activities between authors and boards heavily depend on authors' interest, classification of boards and board popularities. It can be said that authors post more articles to which have more articles or people concerned about.

Observing the principle of these two models and properties in real bipartite network, we propose a new model by using mathematical mechanism. There are two conceptions for this model. First, ways to derive our model is to sample the bipartite network uniformly and use preferential attachment principle. Second, we consider the behaviors in real bipartite social network including one's interest and group effect. The scenario of this model is as follows.

There are $n$ top nodes, $m$ bottom nodes and top nodes are grouped into $b$ groups. For each iteration, a bottom node entered the network and makes $k$ links to top nodes. The ways of choosing top nodes included several modes named $\mathrm{U}, \mathrm{R}, \mathrm{UP_g}, \mathrm{UR}, \mathrm{RP_g}$.

### 2.2 Notations

In this paper, we want to analyze what top node degree will be after a period of time. We want to know the expected degree $E(D_j^t)$ of a top node $j$ after $t$ iterations. At this section, we give notations for this different modes and expected function $E(D_j^t)$.

- $P_{sg}$: probability that the first link and the other links are in the same group.
- $g(j)$: the group which top node $j$ belongs to.
- $W_i$: the weight of group $i$.
- $d_j^t$: degree of top node $j$ at iteration $t$.
- $p$: the number of links at each iteration.

We considered the following modes.

- U : the top nodes are chosen uniformly and independently.
- R : a top node is chosen with probability depending on the degree of it. That is, the probability of node $j$ is $\frac{d_j}{\sum_{k=1}^{n} d_k}$.
- $\mathrm{UP_g}$ : the links made by a bottom node are of two kinds: the first link according to mode U and the remaining $p-1$ links of mode $\mathrm{P_g}$. We defined $\mathrm{P_g}$ later.
- UR : the links made by a bottom node are of two kinds: the first link according to mode $U$ and the remaining $k$-1 links according to mode R.
- $\mathrm{RP_g}$ : the links made by a bottom node are of two kinds: the first link according to mode $R$ and the remaining $k$-1 links according mode $\mathrm{P_g}$.

We also use the following notations.

- $P[i, j]$: a two dimensional probability array. It is the probability of linking to node $j$ in the condition that the first link is to node $i$.
- $P_i(j)$: probability of linking to node $j$ at the $i$th link of a bottom node.
- $P_1^X(j)$: probability of linking to node $j$ at the first link under mode X, $\mathrm{X} \in \{\mathrm{U}, \mathrm{R}\}$ .
- $P^{(\mathrm{XP_g})}[i, j]$, probability of linking to $i$ under mode X and the remain links to node $j$ under mode $\mathrm{P_g}$.

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- X: mode of first link. $\mathrm{X} \in \{\mathrm{U}, \mathrm{R}\}$.
- $\mathrm{P_g}$: mode of remain links, $P_g$ is defined by $P[i,j]$.

# 3 Grouped-RBG

Giving the definition above and then we propose the new bipartite graph model. The new model is as follows.

Fix $n$ top nodes, $m$ bottom nodes and top nodes are grouped into $b$ groups. For each iteration, a bottom node enters the network and makes $p$ links to top nodes. The links made by a bottom node are of two kinds, the first link and the remaining $k-1$ links. However, the $k-1$ links are independent with each other and only depend on the first link.

As described above, we devide this model in different modes, UR, $\mathrm{UP_g}$, $\mathrm{RP_g}$.

$$P[i,j] = \begin{cases} P_{sg} \cdot W_{g(j)} \cdot d_j & \text{if } g(i) = g(j) \\ (1 - P_{sg}) \cdot W_{g(j)} \cdot d_j, & \text{if } g(i) \neq g(j) \end{cases}$$

Y is the expected value of the increased degree of top node $j$ at iteration $t$. It depends on mode X and $\mathrm{P_g}$. Then, the expected degree $E(D_j^t)$ of top node $j$ can be described as follows.

$$E(D_j^{t+1}) = E(D_j^t) + Y$$

Let $D_j^t = E(d_j^t)$.

$$\begin{aligned} Y &= \sum_{p=1}^{k} P_p(j) \\ &= P_1^{\mathrm{X}}(j) + (k-1) \sum_{\forall i} P_1^{\mathrm{X}}(i) P^{(\mathrm{XP_g})}[i,j] \\ &= P_1^{\mathrm{X}}(j) \\ &\quad + \Bigg( \sum_{i \in g(j), i \neq j} P_1^{\mathrm{X}}(i) P^{(\mathrm{XP_g})}[i,j] \\ &\quad + \sum_{i \notin g(j)} P_1^{\mathrm{X}}(i) P^{(\mathrm{XP_g})}[i,j] \Bigg) (k-1) \end{aligned}$$

To predict the features of networks, we use computer programs to simulate the network growths in different scenarios. We show the experiment results at next section.

# 4 Experiments

## 4.1 Experimental results

In this section, we present experiment results under different modes to investigate what the top node degree will be after a period of time. Our major concern is how the different initial conditions affect the final status.

Before the experiments, we check how many iterations the network enters stable state for the modes we defined. We recorded the degree distribution for every 1000 iterations, and compute the maximum difference over all degrees. The result is shown in Table 1. It shows that 10000 iterations are enough.

A scale-free network is a network whose degree distribution follows a power law distribution, at least asymptotically. That is, the fraction of node, $P(k)$, has $k$ connections to other nodes obeys power law, $P(k) \sim ck^{-\gamma}$, in which $c$ is a normalization constant and $\gamma$ is a parameter whose value is typically in the range $2 < \gamma < 3$. As we can see in Figure 1, the distribution shaped as a power-law distribution. The degree distribution of three modes which we define follows the property of scale-free network [2].

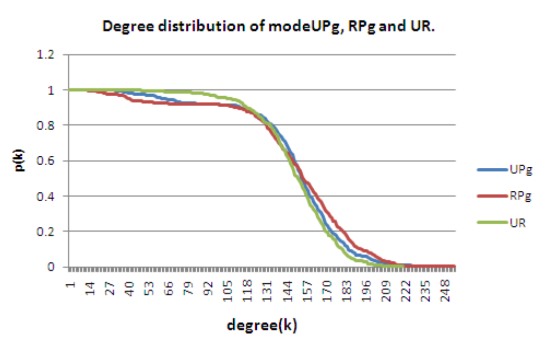


Figure 1: top node degree distribution under modes UR, UPg, RPg

Next, we set different parameters to different modes and analyze the experiment results. We do many experiments for different parameters $P_{sg}$, initial degree of top nodes, the number of groups and other parameters. In this subsection, we show the experiment results and we discuss the results at the next sub-section.

Our experiments are of two kinds.

- Type 1 : For mode $\mathrm{RP_g}$, different initial total degrees of groups.

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- We want to know the difference between top nodes in different groups whose initial degree are lower than others, named $\mathrm{gl}_i$ from the one in group $i$. We give the same initial degree to them and each group has different deviation . The results are shown as Tables 2 and 3, in which $\mathrm{g}_1 = \{100 \times 10,\ 20 \times 10\}$ means that the initial degrees of nodes in group 1 are: 10 top nodes of degree 100 and 10 top nodes of degree20. Also, we check the top nodes' degrees of 2 groups and 4 groups in Figures 2 to 7.
- We compare two other modes UR and $\mathrm{UP_g}$, in Table 4.

• Type 2: The same total initial degrees of groups.

- We observe the group effect in the case that the total degrees are the same but with different groups standard deviation. We set 4 groups, different $\mathrm{P_{sg}}$, different initial conditions, and the results are shown in Table 5.
- We observe the effect of group weight. We compare two groups in different proportion under different $\mathrm{P_{sg}}$. The result is shown as Table 6.

Table 1: Checking convergence for each 1000 bottom nodes.

| convergence test | | | |
|---|---|---|---|
| iterations | UR | $\mathrm{UP_g}$ | $\mathrm{RP_g}$ |
| 1000 | 0.442 | 0.648 | 0.548 |
| 2000 | 0.104 | 0.11 | 0.108 |
| 3000 | 0.052 | 0.046 | 0.046 |
| 4000 | 0.032 | 0.03 | 0.03 |
| 5000 | 0.02 | 0.022 | 0.022 |
| 6000 | 0.018 | 0.016 | 0.018 |
| 7000 | 0.014 | 0.012 | 0.016 |
| 8000 | 0.014 | 0.012 | 0.016 |
| 9000 | 0.014 | 0.01 | 0.01 |
| 10000 | 0.0128 | 0.008 | 0.01 |

## 4.2 Discussion

First, we discuss the results of type 1 experiments. For two groups (Table 2), the total degrees of all $P_{sg}$ in group 1 are all more than group 2 under mode $\mathrm{RP_g}$. The reason is that the initial total degrees of group1 is more than group 2.

Table 2: Total degrees of two groups. $\mathrm{g}_1 = \{100 \times 10,\ 20 \times 10\}$, $\mathrm{g}_2 = \{60 \times 10,\ 20 \times 10\}$.

| | $Psg$ | $\mathrm{g}_1$ | $\mathrm{g}_2$ | $\mathrm{gl}_1$ | $\mathrm{gl}_2$ |
|---|---|---|---|---|---|
| init | | 1200 | 800 | 200 | 200 |
| R | | 37530 | 24470 | 5809 | 5558 |
| RG | 0.8 | 44824 | 17176 | 7074 | 5016 |
| | 0.7 | 44284 | 17716 | 7539 | 4205 |
| | 0.6 | 41034 | 20966 | 7591 | 5102 |
| | 0.5 | 37329 | 24671 | 6389 | 5737 |
| | 0.4 | 34453 | 27547 | 6683 | 6551 |
| | 0.3 | 32381 | 29619 | 5705 | 7010 |
| | 0.2 | 31483 | 30517 | 5116 | 7578 |
| | 0.1 | 31010 | 30990 | 4629 | 7804 |

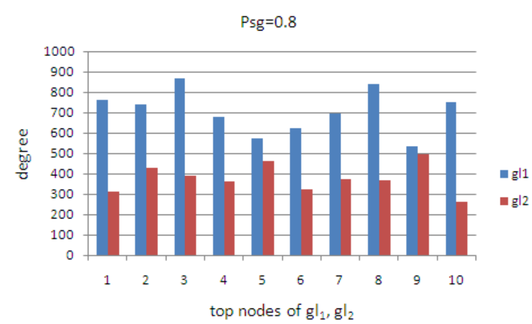


Figure 2: two groups, Psg=0.8

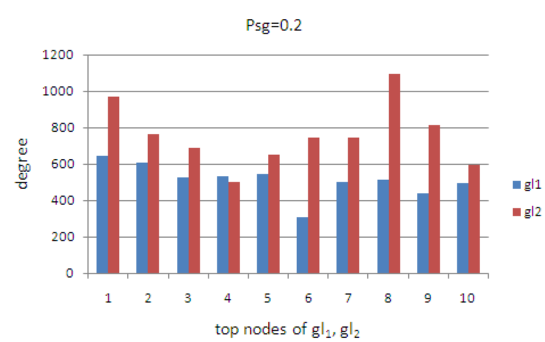


Figure 3: two groups, Psg=0.2

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Table 3: Total degrees of four groups. $g_1 = \{150 \times 15,\ 20 \times 15\}$, $g_2 = \{100 \times 15,\ 20 \times 15\}$, $g_3 = \{80 \times 15,\ 20 \times 15\}$, $g_4 = \{40 \times 15,\ 20 \times 15\}$

| | $Psg$ | g1 | g2 | g3 | g4 | gl1 | gl2 | gl3 | gl4 |
|---|---|---|---|---|---|---|---|---|---|
| init | | 2550 | 1800 | 1500 | 900 | 300 | 300 | 300 | 300 |
| R | | 25505 | 17846 | 14881 | 8518 | 2940 | 2925 | 2975 | 2623 |
| RG | 0.8 | 30008 | 17021 | 12691 | 5376 | 3677 | 3287 | 2307 | 1592 |
| | 0.7 | 27832 | 17263 | 13345 | 6134 | 3491 | 2706 | 2465 | 1867 |
| | 0.6 | 27832 | 17702 | 14133 | 7083 | 3442 | 3003 | 2721 | 2287 |
| | 0.5 | 25797 | 17901 | 14919 | 8133 | 3181 | 3225 | 2823 | 2700 |
| | 0.4 | 24059 | 17850 | 15658 | 9183 | 3014 | 3003 | 2852 | 2938 |
| | 0.3 | 22669 | 18065 | 16013 | 10003 | 2878 | 3125 | 2933 | 3248 |
| | 0.2 | 21622 | 18002 | 16481 | 10645 | 2614 | 3108 | 3117 | 3499 |
| | 0.1 | 20790 | 17955 | 16732 | 11273 | 2422 | 2991 | 3243 | 3691 |

Table 4: The average degree and deviation($\sigma$) of four groups under mode UR and UPg

| | Psg | g1 | g2 | g3 | g4 | g1($\sigma \times 1000$) | g2 | g3 | g4 |
|---|---|---|---|---|---|---|---|---|---|
| init | | 66.667 | 66.667 | 66.667 | 66.667 | 14.731 | .1179 | .1179 | .1179 |
| UR | | 1064.87 | 1061.2 | 1099.2 | 1041.4 | .923071 | .138118 | .208486 | .163575 |
| RPg | 0.8 | 1073.73 | 1051.53 | 1084.67 | 1056.73 | .918206 | .159308 | .179321 | .12947 |

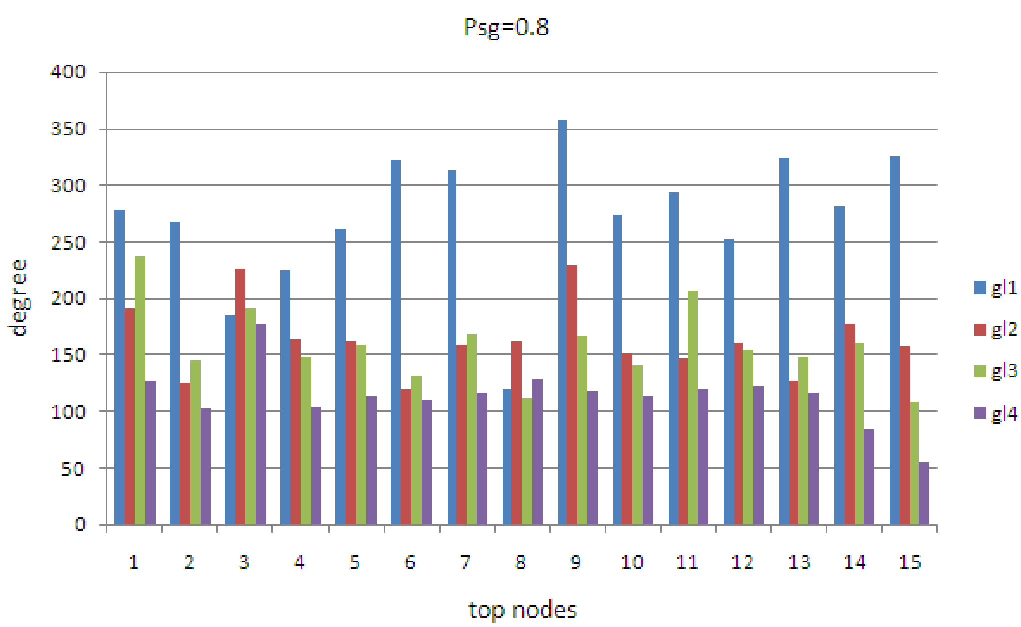


Figure 5: four groups, Psg=0.8,

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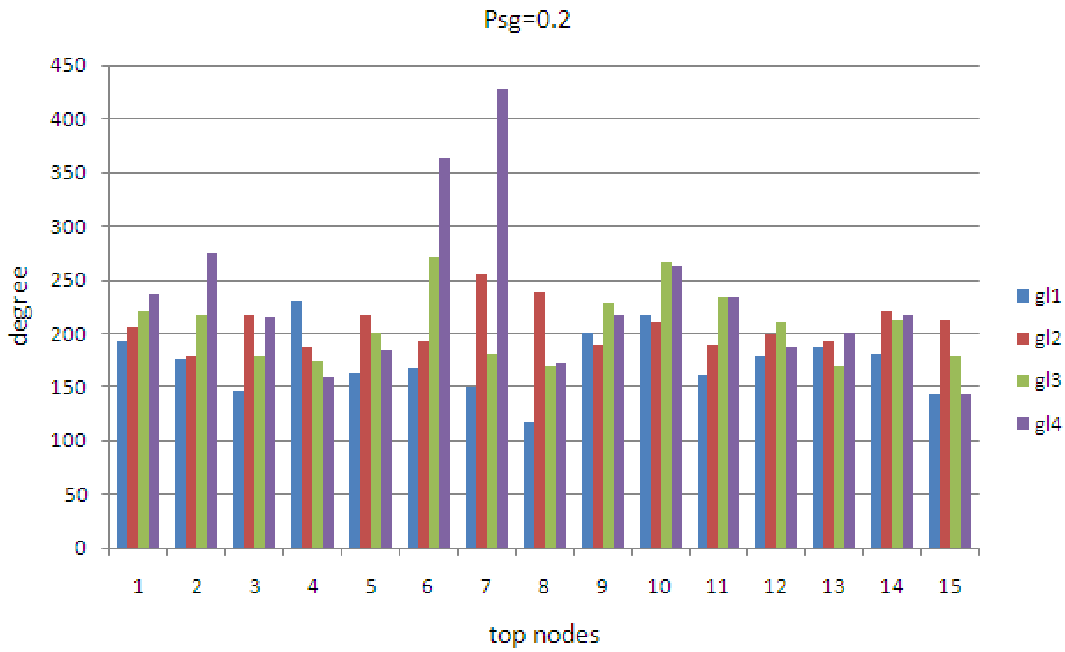


Figure 6: four groups, Psg=0.2

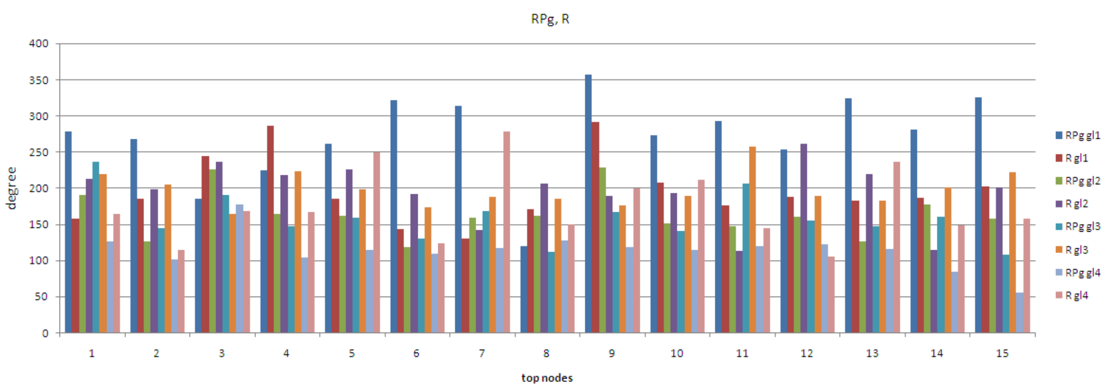


Figure 7: four groups, mode RPg ,Psg=0.8 and mode R

Table 5: The same initial total degree($d$) of four groups. $g_1 = 2$ means 2 top nodes in group 1.

| | $Psg$ | d | $g_1 = 2$ | $g_1 = 3$ | $g_1 = 6$ | $g_1 = 12$ |
|---|---|---|---|---|---|---|
| R | | 600 | 16675 | 16272 | 16016 | 13437 |
| | | 1200 | 16886 | 16892 | 26575 | 14447 |
| | | 2400 | 17823 | 17877 | 17873 | 16027 |
| RPg | 0.8 | 600 | 28055 | 15111 | 15100 | 14134 |
| | | 1200 | 17650 | 15974 | 16241 | 14935 |
| | | 2400 | 18155 | 17424 | 17638 | 16383 |
| | 0.5 | 600 | 16675 | 16272 | 16012 | 13441 |
| | | 1200 | 16717 | 16837 | 16741 | 14502 |
| | | 2400 | 17823 | 17881 | 17869 | 16027 |
| | 0.2 | 600 | 16388 | 15834 | 15807 | 14371 |
| | | 1200 | 16725 | 16479 | 16466 | 15129 |
| | | 2400 | 17802 | 17643 | 17627 | 16528 |

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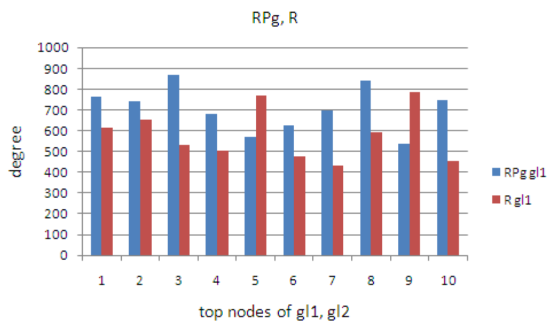


Figure 4: two groups, mode RPg ,Psg=0.8 and mode R

Table 6: Group weight effect under mode $RP_g$. 9v1 means the weight of group 1 and group 2 is 9 to 1.

| Psg | 0.8 | 0.8 | 0.2 | 0.2 |
|---|---|---|---|---|
| | $g_1$ | $g_2$ | $g_1$ | $g_2$ |
| 9v1 | 59615 | 3385 | 56345 | 6655 |
| 8v2 | 57585 | 5415 | 49290 | 13710 |
| 7v3 | 53636 | 9364 | 41215 | 21785 |
| 6v4 | 35917 | 27083 | 33951 | 29049 |
| 5v5 | 31956 | 31044 | 31758 | 31242 |
| 4v6 | 27894 | 35106 | 19591 | 43409 |
| 3v7 | 22520 | 40480 | 10582 | 52418 |
| 2v8 | 14769 | 48231 | 6053 | 56947 |
| 1v9 | 6932 | 56068 | 3790 | 59210 |

However, what we concerned about is $gl_i$, the set of top nodes with small initial degrees such as the 10 top nodes of degree 20 in Table 2. That is, what will the different initial degree of higher ones and $P_{sg}$ affect $gl_i$ in different groups?

The answer is positive. As shown in Table 2, the total degree of $gl_1$ is more than $gl_2$ when $P_{sg}$ is more than 0.5. The reason is that when $P_{sg}$ is more than 0.5, the probability of linking to the same group of first one is more than others. Besides the first link follows mode R. Top nodes which are in the same group of first link will get more chance to be linked by bottom nodes. We check the degree of each top node in $gl_1$ and $gl_2$ in Figure 2 and Figure 3. The degree of top nodes in $gl_1$ are almost more than top nodes in $gl_2$. Therefore, $P_{sg}$ and initial degree of top nodes significantly influence the final state.

And we compare the total degrees of top node in $gl_1$ and $gl_2$ under different modes, $RP_g$ and R. As shown in Table 2, when $P_{sg}$ is more than 0.5, the total degree of group1 under $RP_g$ is more than the one under R. And the result of Figure 4 shows that the degrees of most of the top nodes in $gl_1$ under RPg when $P_{sg} = 0.8$ are more than under R. We may conclude that grouping has great effect in the mode we defined, the small ones will be affected by the larger ones in the same group.

Also we check the result for four groups in Table 3 and from Figure 5 to Figure 7. The above results of two groups also happen here— the total degree in $gl_1$ of larger $P_{sg}$ is more than others. The larger total initial degree of top nodes will highly affect the lower ones of $gl_1$. And the total degree of $gl_1$ under mode RPg is more than under R.

We compare the mode UR and UPg in Table 5, the result shows that the grouping seems have no obvious effect for the top nodes. The reason is that the first link of these two modes links uniformly.

Second, we discuss the result of type 2: the same total initial degree of groups. As the definitions of R and RPg,the top nodes are chosen by their degrees. It seems that the total degree of each group will be the same when the initial total degree are the same. However, the result in Table 4 shows that grouping affects the final result. For the larger $P_{sg}$, the groups of larger initial degree will get more links. And for smaller $P_{sg}$, the total degree of group with larger initial degree of top nodes is also more than others.

Next, we compare the other parameter, group weight. As shown in Table 5, group weight is more effective than $P_{sg}$. No matter how smaller the $P_{sg}$ is, group of larger group weight will get more links

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than other groups.

## 5 Conclusion

In this paper, we propose a new random bipartite graph model, group-RBG. We define three modes $\mathrm{UR}, \mathrm{UP_g}$ and $\mathrm{RP_g}$ of this model with grouping, which happen in real bipartite social networks. We find the typical property, the power-law of the degree distribution, in these modes we defined.

Then we set some different parameters in different scenarios and simulate the bipartite social network. We divided the experiment into two types and we compare the difference between the previous models and ours with grouping. And we realize that grouping is an important factor through the experiment results.

The main result of this paper is that grouping effect in social network important. And it is affected by the top nodes with higher initial degree in the same group. It is interesting in the real social network that a famous member can pull up the popularity of the ones with less popularity in his/her group.

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